

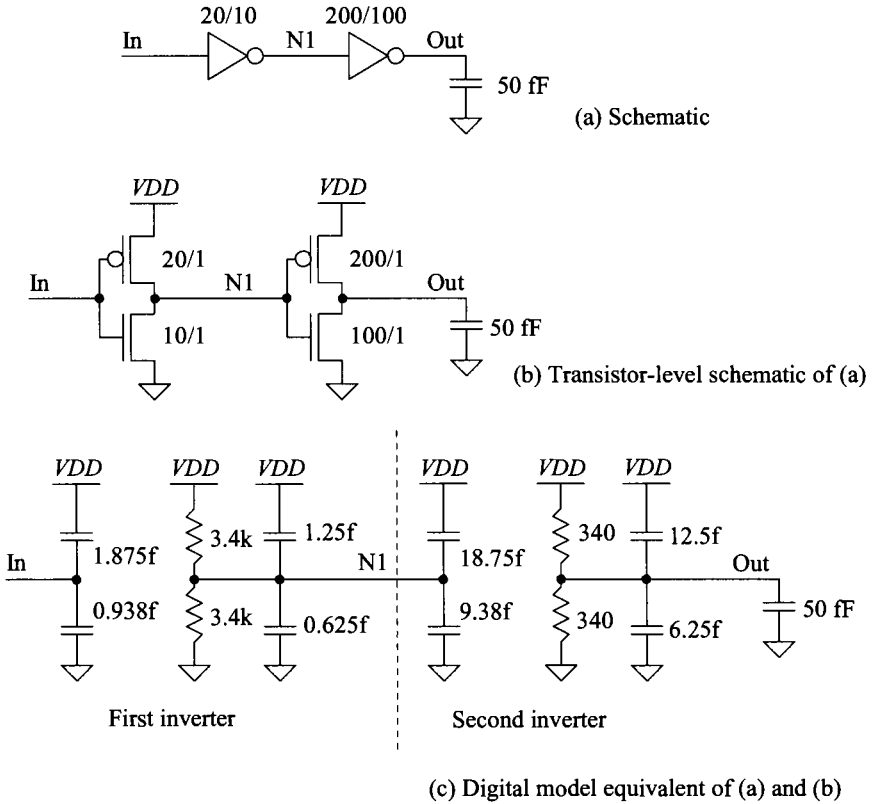


Figure 13.27 Circuit used in Ex. 13.3.

The delay through the second stage, that is, from N1 to the output is calculated as

$$t_{PHL2} + t_{PLH2} = 0.7 \cdot (340 + 340) \cdot 68.75 \text{ fF} = 33 \text{ ps}$$

Again as seen in the first inverter, the effective switching resistance of the PMOS is equal to the NMOS device's effective resistance so $t_{PHL2} = t_{PLH2} = 16.5 \text{ ps}$.

The overall delay through both inverters is

$$t_{PHL} = t_{PLH} = 88 \text{ ps}$$

Figure 13.28 shows the SPICE simulation results. The delays, from the simulations, are approximately 120 ps.

To decrease the delay through the circuit, the simplest solution is to increase the widths of the first inverter. Of course, this increases the circuit's input capacitance. Note that the *analysis* of this circuit has nothing to do with the *design* equations used for implementing a buffer, Sec. 11.4. ■

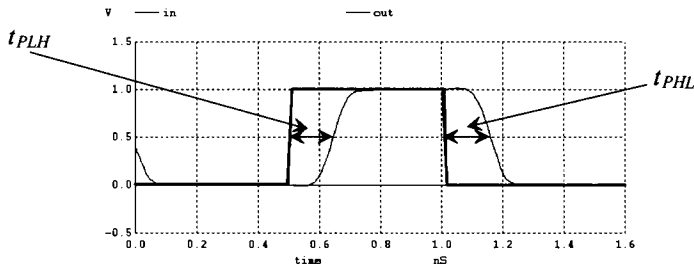


Figure 13.28 Simulating the circuit in Ex. 13.3 and Fig. 13.27.

Example 13.4

In Ex. 13.3 the devices were sized to give (ideally) equal propagation delays. Repeat Ex. 13.3 for the circuit seen in Fig. 13.29.

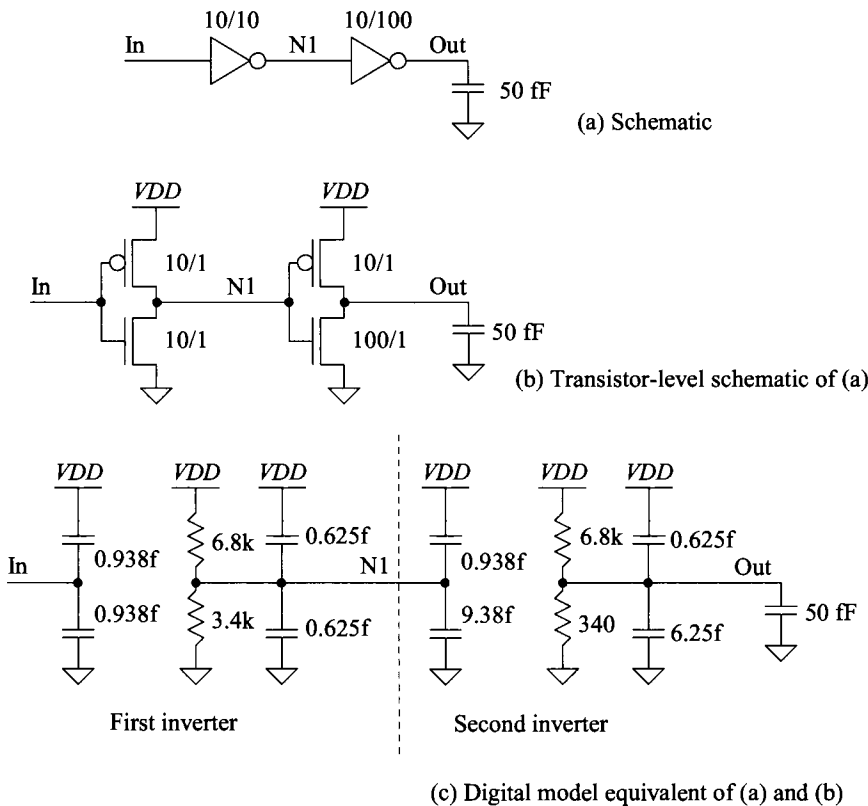


Figure 13.29 Circuit used in Ex. 13.4.

The PMOS device has a width of 10 which is half of the width used for the data in Table 10.2. We can write the effective switching resistance and oxide capacitance for this device as

$$R_p = 6.8k \text{ and } C_{oxp} = 0.625 \text{ fF}$$

As seen in Fig. 13.29, the input capacitance of the circuit is 1.876 fF.

Calculating t_{PLH} begins by noting that when the input goes high the NMOS device in the first inverter turns on. This turns the PMOS device in the second inverter on. The overall delay is calculated as

$$t_{PLH} = \overbrace{0.7 \cdot 3.4k \cdot 11.57 \text{ fF}}^{\text{First inverter's delay}} + \overbrace{0.7 \cdot 6.8k \cdot 56.88 \text{ fF}}^{\text{Second inverter's delay}} \approx 300 \text{ ps}$$

Similarly, when the input goes low, the PMOS in the first inverter turns on and the NMOS in the second inverter turns on, pulling the output low

$$t_{PHL} = \overbrace{0.7 \cdot 6.8k \cdot 11.57 \text{ fF}}^{\text{First inverter's delay}} + \overbrace{0.7 \cdot 340 \cdot 56.88 \text{ fF}}^{\text{Second inverter's delay}} \approx 70 \text{ ps}$$

The simulation results are seen in Fig. 13.30. ■

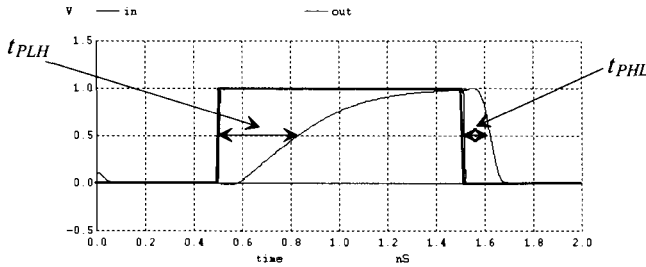


Figure 13.30 Simulating the operation of the circuit in Fig. 13.29.

Example 13.5

Estimate the delay through the circuit seen in Fig. 13.31 using 20/1 PMOS and 10/1 NMOS devices.

Since the load capacitance, 50 fF, is much larger than the output capacitance of the NAND gate, we don't include the NAND gate's capacitance contributions in the digital model seen in Fig. 13.31c. When the input goes high, the NMOS device in the inverter turns on and the PMOS device in the NAND gate turns on (causing the output to go high). The low-to-high delay time is calculated as

$$t_{PLH} = 0.7 \cdot 3.4k \cdot 4.7f + 0.7 \cdot 3.4k \cdot 50f = 130 \text{ ps}$$

The high-to-low delay time is calculated, noting the two NMOS in series when pulling the output low, as

$$t_{PHL} = 0.7 \cdot 3.4k \cdot 4.7f + 0.7 \cdot 6.8k \cdot 50f = 250 \text{ ps}$$

Figure 13.32 shows the simulation results. ■

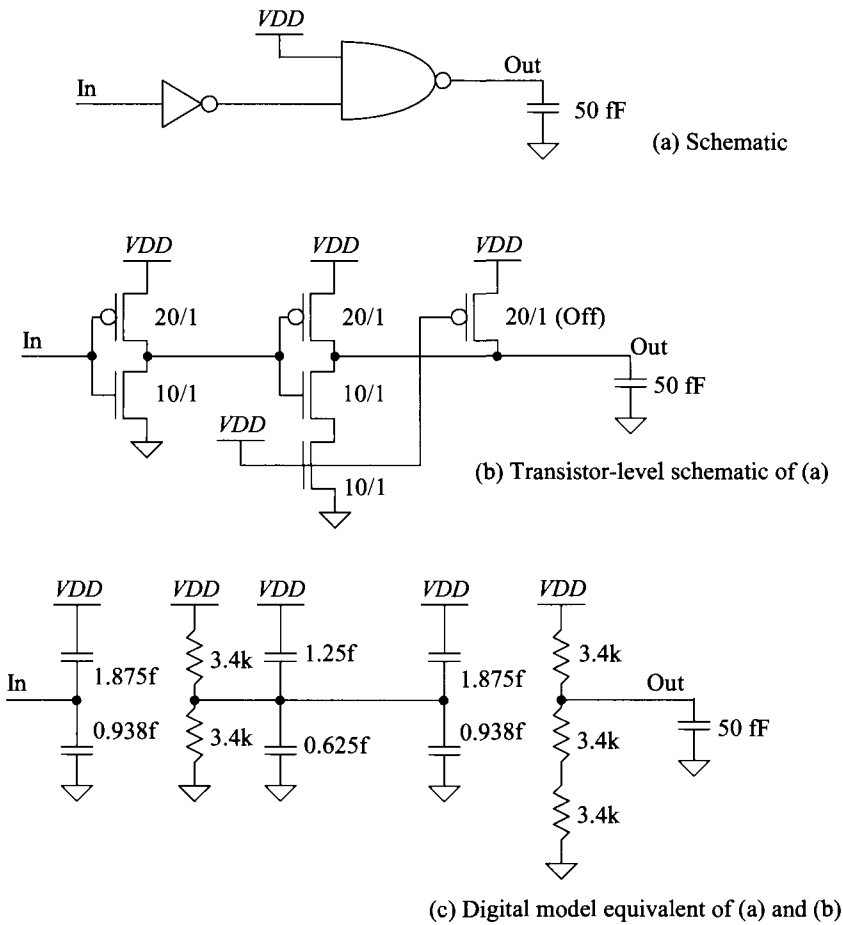


Figure 13.31 Circuit used in Ex. 13.5.

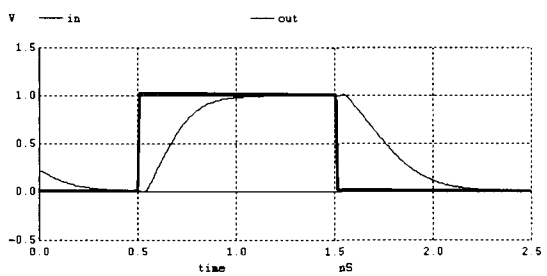


Figure 13.32 Simulating the circuit in Fig. 13.31.

Example 13.6

The circuit seen in Fig. 13.33 is the input portion of the edge triggered D-FF seen in Fig. 13.22. Estimate the delay from the D input to points A and B. Compare the estimate to SPICE simulations. Note that this delay is important because it directly determines the setup time of the FF.

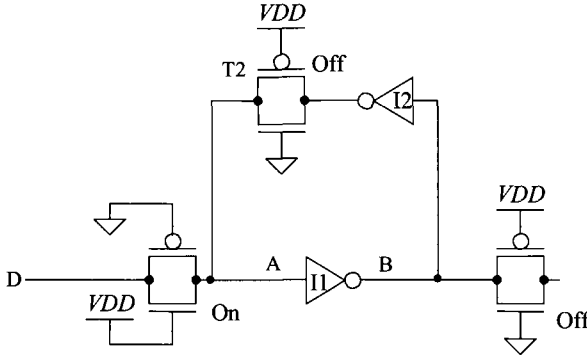


Figure 13.33 Section of a D-FF used in Ex. 13.5.

Node A in Fig. 13.33 is driven through the on TG. The resistance of the TG is $R_n || R_p$. The capacitance on node A is

$$C_A = \underbrace{\frac{C_{oxn}}{2} + \frac{C_{oxp}}{2}}_{\text{Capacitance from the on TG}} + \underbrace{\frac{3}{2} \cdot (C_{oxn} + C_{oxp})}_{\text{Input capacitance of the inverter}} = 3.75 \text{ fF}$$

The first term is the capacitance on node A due to the on TG (see Fig. 10.7). The second term is the input capacitance of the inverter. Note that we have not included the capacitive loading from the off TG. This causes the actual delays to be longer than what we calculate here. (A good exercise at this point is to estimate the capacitive loading of the TGs and include them in the calculations here.) The capacitance on node B is made up of an inverter input capacitance and an inverter output capacitance or

$$C_B = \underbrace{\frac{3}{2} \cdot (C_{oxn} + C_{oxp})}_{\text{Inverter input capacitance}} + \underbrace{(C_{oxn} + C_{oxp})}_{\text{Inverter output capacitance}} = 4.7 \text{ fF}$$

The delay from the D input to point A is estimated as

$$t_{PLHA} = t_{PHLA} = 0.7 \cdot R_n || R_p \cdot C_A = 4.5 \text{ ps}$$

The delay from point A to point B through the inverter is

$$t_{PLHB} + t_{PHLB} = 0.7 \cdot (R_n + R_p) \cdot C_B = 22.5 \text{ ps}$$

The propagation delay from D to B is then

$$t_{PLH} = t_{PHL} = 15.75 \text{ ps}$$

The simulation results are seen in Fig. 13.34. The delay to point A is approximately 30 ps, while the delay to point B is about 90 ps. It's important to reiterate the importance of simulations when determining delays. Hand calculations, as this example shows, have limitations. Hand calculations are still important, however, because they reveal the location of the dominant delays in a digital circuit. ■

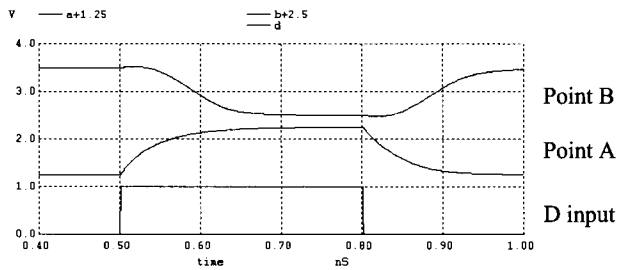


Figure 13.34 Simulating the delays through the latch seen in Fig. 13.33.

ADDITIONAL READING

- [1] M. I. Elmasry, *Digital MOS Integrated Circuits II*, IEEE Press, 1992. ISBN 0-87942-275-0, IEEE order number: PC0269-1.
- [2] J. P. Uyemura, *Circuit Design for Digital CMOS VLSI*, Kluwer Academic Publishers, 1992.
- [3] M. Shoji, *CMOS Digital Circuit Technology*, Prentice-Hall, 1988. ISBN 0-13-138850-9.

PROBLEMS

Unless otherwise stated, use the 50 nm, short-channel CMOS process with the parameters seen in Table 10.2.

- 13.1** Estimate and simulate the delay through 10 TGs connected a 50 fF load capacitance.
- 13.2** Design and simulate the operation of a half adder circuit using TGs.
- 13.3** Sketch the schematic of an 8-to-1 DEMUX using NMOS PGs. Estimate the delay through the DEMUX when the output is connected to a 50 fF load capacitance.
- 13.4** Verify, using SPICE, that the circuit seen in Fig. 13.12 operates as an XOR gate.
- 13.5** Simulate the operation of an SR latch made with NAND gates. Show all four possible input logic combinations.
- 13.6** Simulate the operation of the arbiter seen in Fig. 13.15. Show how two inputs arriving at nearly the same time results in only one output going high.

- 13.7** Show, using simulations, how making the feedback inverter, I2, in Fig. 13.18 stronger (decrease the lengths) can result in the output having either a long delay or not fully switching.
- 13.8** Redesign the FF in Fig. 13.22 without T2 and T4 present. Simulate the operation of your design. Show, by using a limited amplitude on the D input, how point B can have a metastability problems.
- 13.9** In your own words describe setup and hold times. Use the D-FF in Fig. 13.22 and simulations to help support your clear descriptions.

Dynamic Logic Gates

Dynamic or clocked logic gates are used to decrease complexity, increase speed, and lower power dissipation. The basic idea behind dynamic logic is to use the capacitive input of the MOSFET to store a charge and thus remember a logic level for use later. Before we start looking into the design of dynamic logic gates, let's discuss leakage current and the design of clock circuits.

14.1 Fundamentals of Dynamic Logic

Consider the NMOS pass gate (PG) driving an inverter, as shown in Fig. 14.1. If we clock the gate of the PG high, the logic level on the input, point A, will be passed to the input of the inverter, point B. If this logic level is a “0,” the input of the inverter will be forced to ground, while a logic “1” will force the input of the inverter to $V_{DD} - V_{THN}$. When the clock signal goes low, the PG shuts off and the input to the inverter “remembers” the logic level. In other words, when the PG turns on, the input capacitance of the inverter is either charged to $V_{DD} - V_{THN}$ or discharged to ground through the PG. As long as this charge is present on the parasitic input capacitance of the inverter, the logic value is remembered. What we are concerned with at this point are the leakage mechanisms which can leak the stored charge off the node. A node, such as the one labeled B in Fig. 14.1, is called a dynamic node or a storage node. Note that this node is a high-impedance node and is easily susceptible to noise (see Ex. 3.5).

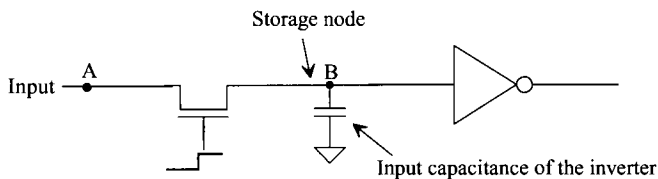


Figure 14.1 Example of a dynamic circuit and associated storage capacitance.

14.1.1 Charge Leakage

One of the important concerns with designing dynamic circuits is the MOSFET's off current, see Sec. 6.4.2. In Fig. 14.2 we show the simulated drain current of a 10/1 MOSFET in the 50 nm process plotted against gate-source voltage. The off current, with $V_{GS} = 0$, is taken from the plot as

$$\log I_D = -8.45 \rightarrow I_D = 3.55 \text{ nA} = I_{off} \cdot W \cdot \text{scale} \quad (14.1)$$

noting that the worst case off current occurs when the minimum channel length is used ($L = 1$). The off current can be estimated as

$$I_{off} = 7.1 \text{ nA}/\mu\text{m} \quad (14.2)$$

The off current is made up of leakage through the source/drain implant to substrate (the formed diode and the associated saturation currents), leakage around the edges of the poly at the edge of the field oxide, and by the source-to-drain leakage currents.

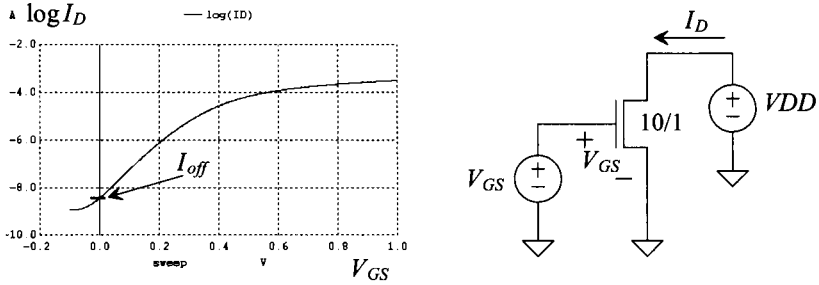


Figure 14.2 Drain current of an NMOS device plotted from weak to strong inversion. See Fig. 6.16. PMOS netlist is found at cmosedu.com.

The rate at which the storage node seen in Fig. 14.1 discharges is given by

$$\frac{dV}{dt} = \frac{I_{off} \cdot W \cdot \text{scale}}{C_{node}} \quad (14.3)$$

The node capacitance, C_{node} , is the sum of the input capacitance of the inverter, the capacitance to ground of the metal or poly line connecting the inverter to the pass transistor, and the capacitance of the drain implant to substrate (the depletion capacitance). For practical applications, we assume that

$$C_{node} \approx C_{in} \text{ of the inverter} \quad (14.4)$$

Example 14.1

Estimate the discharge rate of the 50 fF capacitor shown Fig. 14.3.

Using Eqs. (14.1) and (14.3), we can write the capacitor's discharge rate as

$$\frac{dV}{dt} = \frac{3.55 \text{ nA}}{50 \text{ fF}} = 71 \text{ mV}/\mu\text{s}$$